

Stable low-dimensional perovskitoids for high-performance all-perovskite tandem solar cells

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A list of authors and their affiliations appears at the end of the paper

Low-dimensional perovskites have been widely explored to enhance the performance of perovskite solar cells. However, conventional low-dimensional perovskites rely on amino-based cations with weak hydrogen bonds, which makes them unstable due to facile deprotonation. Here we introduce stable trimethylphosphonium-based cations featuring P-centred ligands with robust $-\text{CH}_3$ groups. These cations exhibit significantly higher acid dissociation constants, suppressed Hofmann elimination and reduced susceptibility to bimolecular nucleophilic substitution, thereby inhibiting degradation while providing strong defect passivation. A single-crystal structural analysis revealed that these cations assemble into a one-dimensional perovskitoid architecture that surrounds and protects the inorganic Pb-I framework, thereby conferring exceptional resistance to moisture, heat and electrical bias. Consequently, single-junction 1.25-eV-bandgap Sn-Pb perovskite solar cells achieved a power conversion efficiency of 24.19%. Moreover, monolithic all-perovskite tandem devices reached a high open-circuit voltage of 2.224 V and a power conversion efficiency of 30.17% (certified 29.18%). These tandem devices retained over 90% of their initial efficiency for more than 2,000 h under continuous maximum power point tracking. Furthermore, the one-dimensional trimethylphosphonium cation-surrounded perovskitoids were used in high-performance 1.77-eV-bandgap perovskite solar cells and green perovskite light-emitting diodes, thus showing that they offer a versatile platform for efficient and stable perovskite optoelectronic devices.

In recent years, the power conversion efficiency (PCE) of both single-junction and tandem perovskite solar cells (PSCs) has increased rapidly, thereby establishing PSCs as leading contenders for next-generation photovoltaic technologies^{1–7}. Achieving higher efficiency remains a central research focus, with bulk additives and surface passivation emerging as highly effective strategies^{8–14}. Among these, amino-based cation additives have been extensively employed to mitigate defects and promote the formation of low-dimensional (LD) perovskites or their derivatives, thereby improving the performance of both single-junction and tandem devices^{6,9,15–17}.

Compared with three-dimensional perovskites, LD perovskites provide distinct advantages, including enhanced defect passivation, increased stability and suppressed ion migration^{18–24}. A wide variety of amino species, from mono- and diamino cations to mixed-cation systems, have been shown to effectively suppress defects and improve carrier transport. For example, an amino-based two-dimensional perovskite layer at the buried interface can promote uniform crystallization and defect passivation, whereas amino-based cations can induce dipolar passivation and enhance ohmic contact, thereby enabling efficient hole extraction in solar cells^{25,26}. Beyond photovoltaics,

✉ e-mail: weijun.ke@whu.edu.cn

amino-based LD perovskites have also found wide use in optoelectronic devices, such as photodetectors and light-emitting diodes^{27,28}.

Despite their advantages, amino-based cations face intrinsic limitations. Their relatively weak hydrogen bonding and low acid dissociation constants (pK_a) restrict both their defect passivation capability and long-term stability enhancement^{18,29,30}. Moreover, conventional one-dimensional (1D) and two-dimensional amino-based LD perovskites only partially protect the inorganic Pb–I framework, leaving many facets exposed and susceptible to environmental degradation^{23,31}. These challenges are even more pronounced in all-perovskite tandem solar cells (TSCs), in which the narrow-bandgap (NBG) mixed Sn–Pb perovskite subcells are particularly prone to oxidation³². Consequently, although certified all-perovskite TSCs have achieved efficiencies up to 30.1% (ref. 26), their operational stability remains inadequate: state-of-the-art >28% devices typically show a noticeable performance decay within ~1,000 h, a critical barrier to future commercialization. Collectively, these limitations underscore the urgent need for new cation designs that transcend conventional amino-based systems and provide comprehensive structural protection and superior intrinsic stability, which would benefit not only single-junction and tandem PSCs but also a wide range of perovskite optoelectronic technologies.

Here we introduce a tailored trimethylphosphonium-based cation, benzyltrimethylphosphine (BTP), featuring a P-centre core and hydrogen-rich $-CH_3$ groups. Substituting N with P and introducing $-CH_3$ substituents markedly increased the pK_a , suppressed Hofmann elimination and reduced susceptibility to bimolecular nucleophilic substitution. As a result, the cation exhibits significantly enhanced resistance to dissociation and provides stronger defect passivation compared with conventional ammonium analogues. Beyond its intrinsic robustness, a single-crystal analysis revealed that BTP assembles into a 1D perovskitoid structure in which the inorganic Pb–I framework is fully shielded by BTP cations, thereby affording exceptional thermal and moisture stability. The incorporation of benzyltrimethylphosphine chloride (BTPCl) into NBG Sn–Pb perovskite films notably enhanced their optoelectronic properties and stability. As a result, single-junction 1.25-eV-bandgap PSCs achieved a PCE of 24.19%. Moreover, all-perovskite tandem devices reached a PCE of 30.17% with a high open-circuit voltage (V_{oc}) of 2.224 V and a time to 90% initial efficiency (T_{90}) lifetime exceeding 2,000 h under continuous maximum power point (MPP) tracking. Furthermore, this strategy improved the performance of 1.77-eV-bandgap PSCs and green perovskite light-emitting diodes (PeLEDs), thereby underscoring the broad potential of 1D trimethylphosphonium cation-surrounded perovskitoids as a robust platform for advanced perovskite optoelectronic technologies.

Selection of trimethylphosphonium cations

Phosphonium salts have recently attracted attention for their roles in interfacial defect passivation³³, the construction of LD semiconductors³⁴ and two-dimensional hybrid ferroelectrics³⁵. These studies highlighted the considerable potential of phosphonium salts in optoelectronic devices. However, their specific role as the primary cations in forming LD perovskites or perovskite-analogue structures for solar cell devices, particularly relative to conventional ammonium analogues, has remained largely unexplored, warranting systematic investigation. At present, the most commonly used molecules for passivation, either as additives or for LD perovskite formation, generally contain ammonium groups^{17,36,37}. However, these groups are intrinsically unstable due to their susceptibility to dissociation under external stimuli^{18,29,38}.

To overcome this intrinsic limitation, we propose a three-step design strategy and ultimately target a phosphorus-centred BTP⁺ cation (Fig. 1a). First, full methylation of the central atom yields a quaternary cationic framework, which substantially increases pK_a and thereby suppresses acid-base degradation (Supplementary Fig. 1), the dominant failure pathway. However, even quaternary ammonium salts (R_4N^+) remain vulnerable to thermal degradation via Hofmann elimination,

which requires the presence of β -hydrogen (β -H) atoms³⁹. In our design, we used benzyl and methyl substituents to eliminate the β -H sites, thereby stoichiometrically blocking this degradation pathway and ensuring thermal robustness during high-temperature processing (Fig. 1a). Another key degradation pathway in quaternary systems is bimolecular nucleophilic substitution (S_N2) at the α -carbon (α -C)⁴⁰. In ammonium salts, the high electronegativity of nitrogen strongly polarizes the C–N bonds, rendering the α -C highly electrophilic and susceptible to nucleophilic attack (for example, by iodide ions)⁴¹. By substituting nitrogen with phosphorus, the electronic structure is fundamentally altered. The lower electronegativity of phosphorus reduces inductive electron withdrawal, whereas its larger, more polarizable electron cloud enables greater charge delocalization. As a result (Supplementary Fig. 2), BTP⁺ exhibits a more diffuse electron distribution, with an isotropic polarizability of 18.78 Å³, 12.2% higher than that of the ammonium analogue benzyltrimethylammonium cation (BTMA⁺, 16.74 Å³). When passivating a formamidinium (FA) vacancy on the perovskite surface, both cations exhibit similarly strong binding energies (E_b = –4.96 eV for BTMA⁺ and –4.98 eV for BTP⁺). However, an energy decomposition analysis revealed a stronger dispersion interaction for BTP⁺ (E_{disp} = –0.79 eV) compared with BTMA⁺ (–0.73 eV), indicating that the phosphorus-based cation compensates for weaker electrostatic interactions through enhanced van der Waals contributions. A similar trend was observed in the comparison between phenylethylammonium ions (PEA⁺) and BTP⁺ (Supplementary Fig. 3). This diffuse electronic structure also suppresses the S_N2 degradation pathway: unlike BTMA⁺, in which polarized methyl groups create electrophilic α -C vulnerable to nucleophilic attack, BTP⁺ exhibits reduced α -C electrophilicity, leading to improved chemical stability.

These design features render the BTP⁺ cation substantially more stable than conventional PEA⁺. A comparative analysis confirmed the superior stability of BTP⁺ relative to PEA⁺ (Fig. 1b,c). The calculated pK_a values were 9.82 for PEA⁺ and 22.97 for BTP⁺, indicating substantially enhanced resistance to deprotonation. Beyond enhanced deprotonation resistance, we also observed that introducing $-CH_3$ groups increased the number of hydrogen atoms, thereby potentially strengthening the organic–inorganic interactions and, thus, stabilizing the perovskite framework. To quantify these interactions, we conducted a Hirshfeld surface analysis, as shown in Fig. 1d. The BTP–[PbI₆] interaction ratio is 34.4%, higher than the 31.1% observed for PEA–[PbI₆], confirming a stronger H–I interaction probability. Additionally, the organic BTP cation effectively isolates the inorganic framework in (BTP)PbI₃ (Fig. 1c), which could enhance humidity resistance⁴².

To evaluate the stability of (BTP)PbI₃ relative to (PEA)₂PbI₄, we performed ab initio molecular dynamics simulations with one water molecule inserted and analysed the structural evolution (Fig. 1e,f). Simulations were conducted at 500 K (the temperature reached equilibrium after 1 ps, as shown in Supplementary Fig. 4) to accelerate structural degradation. Snapshots before and after structural relaxations can be found in Supplementary Fig. 5. For (PEA)₂PbI₄, a reversible deprotonation–protonation process occurred at 1.54 ps (Fig. 1e): one PEA⁺ cation loses a proton, which is captured by H₂O to form a H₃O⁺ cation and neutral PEA⁰. Over time, PEA⁰ regains a proton from H₃O⁺, thereby reforming the PEA⁺ and H₂O and ultimately leading to hydrogen bonding between PEA⁺ and H₂O (bond length of 1.72 Å). This behaviour highlights the susceptibility of PEA⁺ cations to deprotonation in humid conditions. By contrast, throughout the simulation, the water molecule in (BTP)PbI₃ remained distant from the surrounding BTP⁺ cations (Fig. 1f), thereby demonstrating its superior moisture resistance as expected. These results indicate that the strong organic–inorganic interactions between the BTP cation and the [PbI₆] framework can reinforce the perovskite structure, in particular, to confer superior moisture stability. Collectively, these structural and electronic advantages establish BTP⁺ as a highly promising cation for constructing stable LD perovskite and perovskitoid systems.

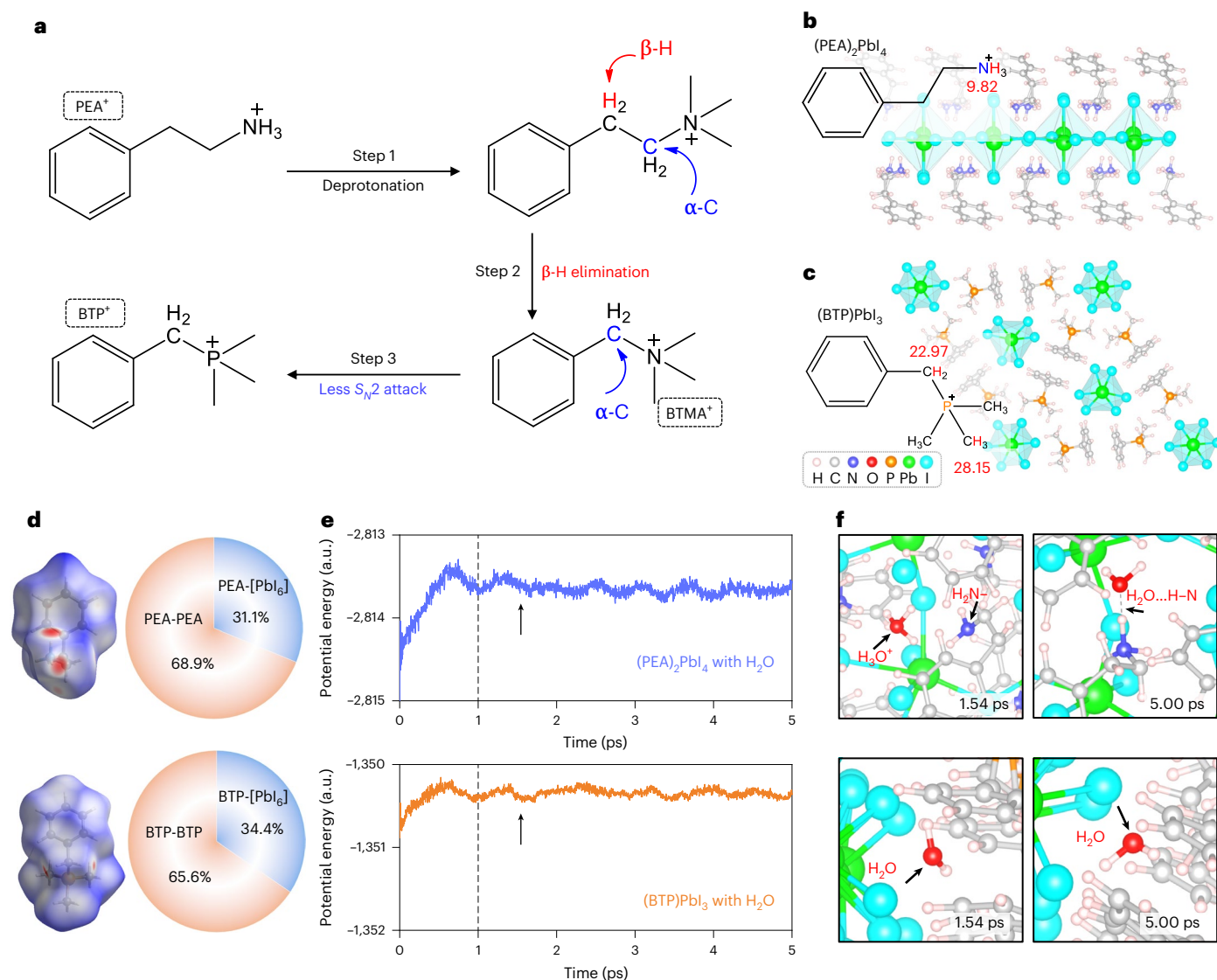


Fig. 1 | Selection of P-based cations. **a**, A three-step design strategy from PEA⁺ to BTP⁺. **b, c**, Comparison of pK_a values for PEA (**b**) and BTP (**c**) in their respective perovskite structures. **d**, Hirshfeld surfaces and relative contributions to the Hirshfeld surface area of various close intermolecular contacts in (PEA)₂PbI₄

and (BTP)PbI₃. **e, f**, Energy profiles simulated by ab initio molecular dynamics of (PEA)₂PbI₄ (**e**, top) and (BTP)PbI₃ (**e**, bottom) and corresponding snapshots of (PEA)₂PbI₄ and (BTP)PbI₃ at 1.54 ps (**f**, left) and 5.00 ps (**f**, right).

One-dimensional cation-surrounded perovskitoids and their influence on stability

Beyond its enhanced resistance to degradation under humid conditions compared with PEA⁺, the BTP cation also differs structurally. Although PEA typically forms a planar-clipped two-dimensional perovskite structure, BTP can form a cation-surrounded 1D perovskitoid structure. As shown in Fig. 2a, the 1D (BTP)PbI₃ structure consists of face-sharing Pb–I octahedra, such that each octahedral chain is fully surrounded by six BTP molecular chains. This arrangement provides robust protection for the Pb–I octahedra, thereby enhancing stability under external stressors. Structural data obtained from a single-crystal analysis are summarized in Supplementary Table 1. The superior stability of (BTP)PbI₃ was confirmed by X-ray diffraction, ultraviolet–visible (UV–vis) absorption spectroscopy and time-of-flight secondary-ion mass spectrometry (TOF-SIMS) depth profiling, which revealed markedly improved structural, optical and compositional stability compared with (PEA)₂PbI₄ (Supplementary Figs. 6–9).

We incorporated BTPCl as an additive into NBG mixed Sn–Pb perovskite films, which are particularly susceptible to degradation under

ambient conditions. A chloride salt (BTPCl; Supplementary Figs. 10 and 11) was selected to exploit the synergistic benefits of both the BTP⁺ cation and the Cl[−] anion^{43,44}. We first examined the moisture stability of the perovskite films. As shown in Fig. 2b, the water contact angle of the control film decreased from 55.1° to 51.6° after 9 s, indicating poor water resistance⁴⁵. By contrast, the BTPCl-incorporated film exhibited the highest water contact angle and the least change over time, decreasing only from 62.7° to 61.8° after 9 s, outperforming both the control and samples with incorporated phenylethylammonium chloride (PEACl). To further evaluate stability, we immersed entire perovskite films in water. As shown in Fig. 2c, the control and PEACl-incorporated films fully decomposed within 10 min and 15 min, respectively. By contrast, the BTPCl-incorporated film remained largely intact, with only minor decomposition at the edges. Increasing the BTPCl concentration further delayed the onset of yellowing (Supplementary Fig. 12). These results confirm that BTPCl markedly enhanced the water stability of perovskite films. An X-ray diffraction analysis and density functional theory calculations revealed that BTPCl does not become incorporated into the three-dimensional Sn–Pb perovskite lattice; instead, it favours

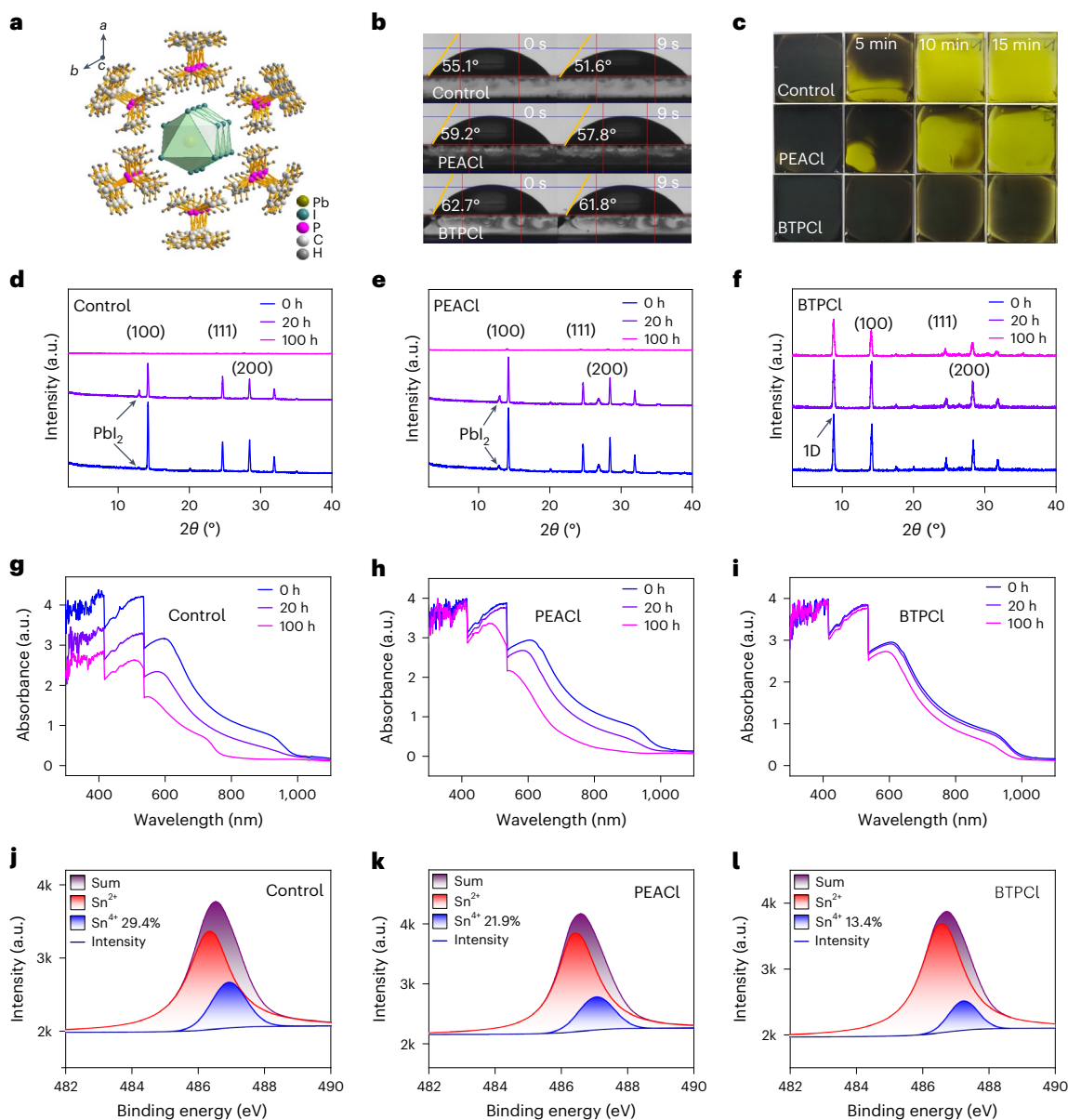


Fig. 2 | One-dimensional structure and its influence on the moisture and thermal stability of perovskite films. **a**, Crystal structure of 1D all-cation-surrounded (BTP)PbI₃. **b**, Measured water contact angles for control, PEACl-treated (10 mg ml⁻¹) and BTPCl-treated (10 mg ml⁻¹) perovskite films, captured at the initial moment and after 9 s in ambient air. **c**, Photographs showing the time-dependent decomposition of control, PEACl-treated and BTPCl-treated perovskite films upon immersion in deionized water. **d–f**, X-ray diffraction

ageing profiles of control (**d**), PEACl-treated (10 mg ml⁻¹) (**e**) and BTPCl-treated (10 mg ml⁻¹) (**f**) perovskite films. **g–i**, UV–vis absorption ageing profiles of a control film (**g**), a PEACl-treated film (10 mg ml⁻¹) (**h**) and a BTPCl-treated film (10 mg ml⁻¹) (**i**) under continuous heating in ambient air at 85 °C. **j–l**, X-ray photoelectron spectroscopy spectra of Sn atoms on the top surface of a control film (**j**), a PEACl-treated film (**k**) and a BTPCl-treated film (**l**).

the formation of a 1D (BTP)PbI₃ perovskitoid structure rather than a ‘three-dimensional’ (BTP)PbI₃ phase (Supplementary Figs. 13 and 14).

Atomic force microscopy and Kelvin probe force microscopy measurements revealed that the BTPCl-incorporated film exhibited the lowest surface roughness and the most stable surface potential (Supplementary Figs. 15 and 16), indicating that BTPCl effectively shielded the perovskite film, thereby mitigating the degradation caused by exposure to environmental factors such as moisture and oxygen. Furthermore, as shown in Fig. 2d–f, after 20 h, the characteristic lead iodide (PbI₂) peak at 12.8° was substantially intensified in the control and PEACl-incorporated films, indicating substantial degradation. After 100 h, the X-ray diffraction patterns of these films became notably weaker and nearly undetectable. By contrast, the BTPCl-incorporated film showed no obvious PbI₂ peak and only

minimal changes in peak intensity, demonstrating its excellent resistance to thermal and moisture-induced degradation. Similarly, UV–vis absorption spectra revealed a noticeable decrease in absorption intensity for the PEACl-incorporated film after prolonged exposure to heat and humidity (85 °C, 65% relative humidity) (Fig. 2g–i). Conversely, the BTPCl-incorporated film maintained relatively stable absorption. The X-ray photoelectron spectroscopy analysis revealed that the BTPCl-treated film exhibited a notably lower Sn⁴⁺ content of around 13.4%, outperforming both the control film (29.4%) and PEACl-treated film (21.9%) (Fig. 2j–l), thereby indicating enhanced resistance to oxidation. This stabilization was also directly observable in precursor solutions exposed to ambient air (Supplementary Fig. 17). Furthermore, BTPCl substantially enhanced the stability of the film under a reverse bias stress (Supplementary Figs. 18 and 19).

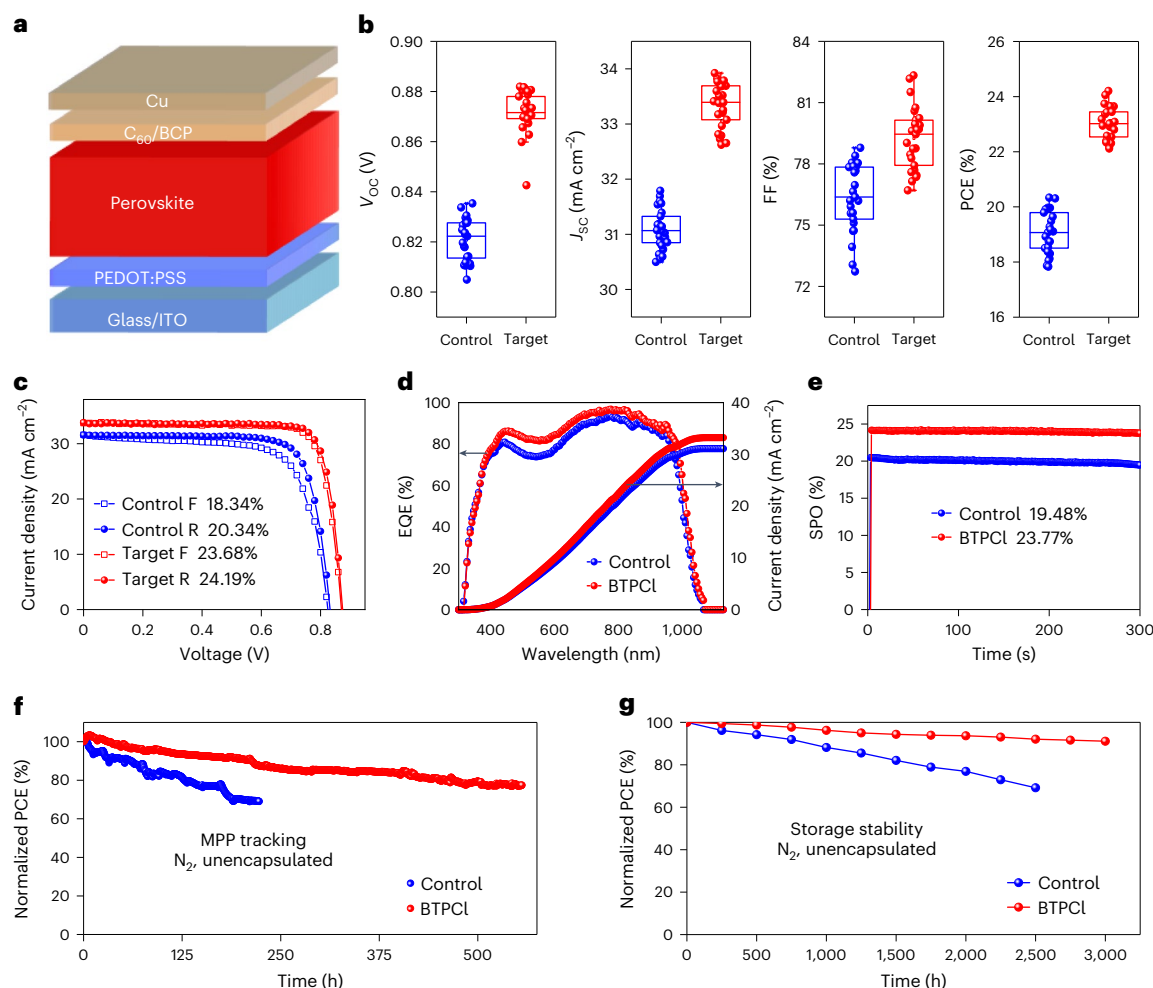


Fig. 3 | Single-junction NBG mixed Sn–Pb PSCs. **a**, Device architecture of single-junction FA_{0.7}MA_{0.3}Pb_{0.5}Sn_{0.5}I₃ PSCs. **b**, Statistical distribution of V_{oc} , J_{sc} , FF and PCE for 25 individual control devices and 25 individual BTPCl-treated mixed Sn–Pb PSCs fabricated in identical runs. The central lines in the box plots represent the medians and reflect the median trends of the data. The width of a box indicates the degree of dispersion in the data. Outliers are represented by points outside this range. **c**, J – V curves of representative control and the BTPCl-incorporated mixed Sn–Pb PSCs measured under both forward and reverse voltage scans. **d**, **e**, EQE spectra (**d**) and SPO efficiencies (**e**) of control

and BTPCl-incorporated mixed Sn–Pb PSCs. **f**, Stability tests of unencapsulated control (initial efficiency 19.85%) and BTPCl-incorporated (initial efficiency 23.16%) mixed Sn–Pb PSCs under MPP tracking with 1-sun irradiation (a light-emitting diode light source, equivalent to AM1.5 G, 100 mW cm⁻²) in an N₂-filled glovebox at approximately 55 °C. Data were normalized to the initial PCE. **g**, Long-term stability of unencapsulated control (initial efficiency 19.37%) and BTPCl-incorporated (initial efficiency 23.34%) mixed Sn–Pb PSCs stored in an N₂ environment at room temperature. F, forward voltage scan; R, reverse voltage scan.

Characteristics of BTPCl-incorporated Sn–Pb perovskite films

The BTPCl cation exhibited strong interactions with the perovskite structure (Supplementary Fig. 20), and 1D (BTP)PbI₃ perovskitoid structures were observed in the final films (Supplementary Figs. 21 and 22). TOF-SIMS measurements confirmed that phosphorus (P⁺), a characteristic element of BTP, was distributed throughout the entire perovskite layer (Supplementary Fig. 23). Moreover, the incorporation of BTP cations reduced the internal stress, thus resulting in uniform morphology and a lower surface roughness (Supplementary Figs. 24–26). In addition, BTPCl-treated perovskite films exhibited a reduced exciton binding energy E_b , enhanced photoluminescence (PL) emission and prolonged carrier lifetime, which collectively confirm the improved crystalline quality and reduced defect density and, thereby, facilitate more efficient carrier transport (Supplementary Figs. 27–31 and Supplementary Table 2). UV photoelectron spectroscopy and Kelvin probe force microscopy further revealed an upward shift in the Fermi level (E_F) and a slight increase in the surface potential upon BTPCl treatment (Supplementary Figs. 32–35),

indicating a more favorable electronic structure that suppresses charge recombination losses^{46,47}.

Single-junction 1.25-eV-bandgap Sn–Pb PSCs

Inverted PSCs with the indium tin oxide (ITO)/PEDOT:PSS/FA_{0.7}MA_{0.3}Pb_{0.5}Sn_{0.5}I₃ (with or without BTPCl)/C₆₀/bathocuproine (BCP)/Cu structure were fabricated, as illustrated in Fig. 3a. Here MA is methylammonium. PEDOT and PSS are defined in Methods. The perovskite layers had a thickness of approximately 950 nm. The grain orientation was improved in the BTPCl-treated films (Supplementary Fig. 36). Compared with the pristine devices, all photovoltaic parameters of the BTPCl-incorporated devices were improved (Fig. 3b and Supplementary Fig. 37). Specifically, the BTPCl devices exhibited an average PCE of 23.4 ± 0.8% across 25 devices. The best-performing device achieved a V_{oc} of 0.87 V, a short-circuit current density (J_{sc}) of 33.68 mA cm⁻², a fill factor (FF) of 82.33% and a PCE of 24.19%, under reverse voltage scanning (Fig. 3c and Supplementary Table 3). Notably, BTPCl substantially improved the current, which was consistent with the external quantum efficiency (EQE) results shown in Fig. 3d. The increased current

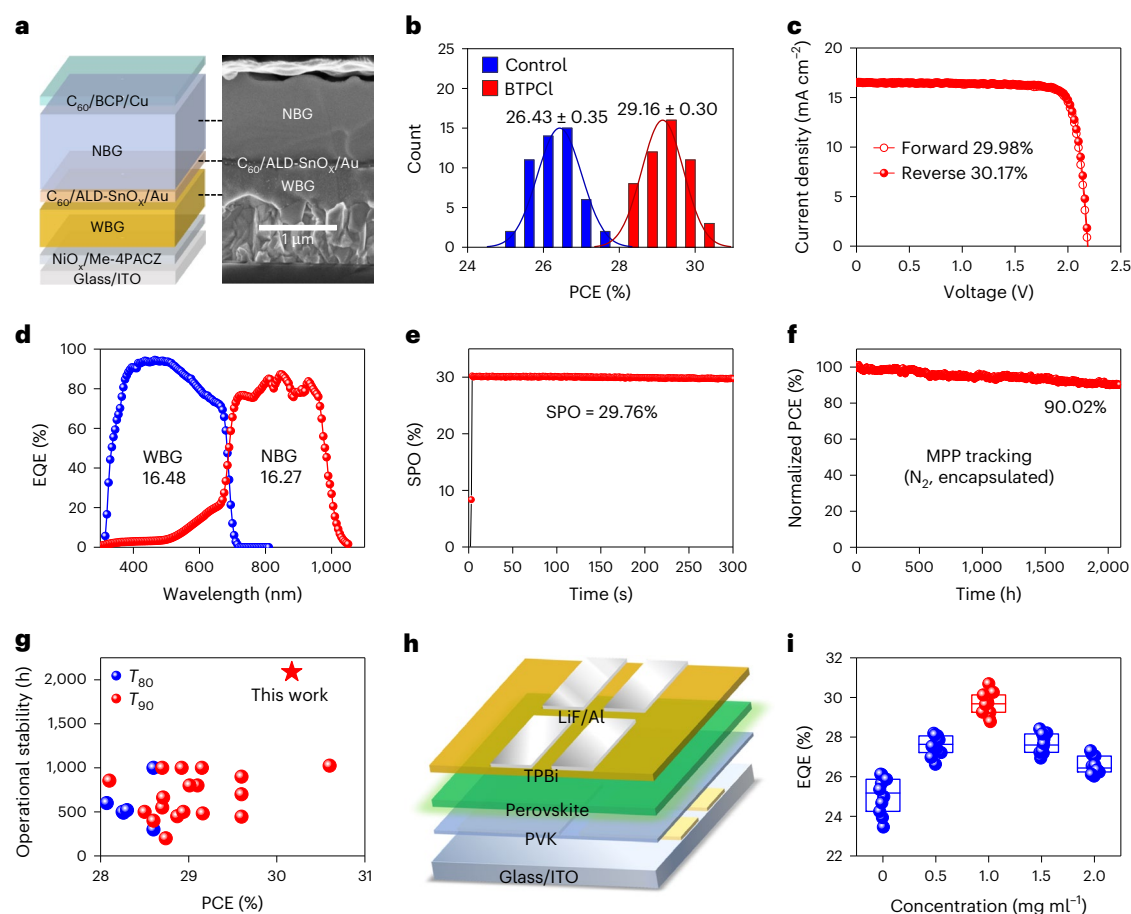


Fig. 4 | All-perovskite TSCs and other optoelectronic devices. **a**, Schematic of the all-perovskite TSC structure and a cross-sectional scanning electron microscopy image of a representative device. **b**, Statistical distribution of PCE for 50 individual control devices and 50 BTPCI-treated all-perovskite TSCs. **c**, $J-V$ curves of the best-performing all-perovskite TSCs. **d**, **e**, EQE spectra (**d**) and SPO efficiency (**e**) of the best-performing tandem device. **f**, Continuous MPP tracking of encapsulated all-perovskite TSCs (initial efficiency 29.53%) under 1-sun irradiation (a light-emitting diode light source, equivalent to AM1.5 G, 100 mW

cm^{-2}) in an N_2 -filled glovebox at $-55^\circ C$. **g**, Summary of the efficiency and stability evolution (with MPP tracking) of all-perovskite TSCs, where T_{90} and T_{80} denote the time to 90% and 80% of their initial PCEs, respectively. **h**, **i**, Device structure (**h**) and statistical distribution of EQE (**i**) for pure green PeLEDs with various BTPCI concentrations (0–2 $mg\ ml^{-1}$). The central line in each box-and-whisker plot represents the median value. The box indicates the interquartile range and, thus, the data dispersion. Points outside this range are defined as outliers.

can be primarily attributed to enhanced light absorption in the films (Supplementary Fig. 38) and improved crystallinity following BTPCI incorporation, even though both films exhibited similar bandgaps of approximately 1.25 eV (Supplementary Fig. 39). Figure 3e shows that the steady-state power output (SPO) efficiency of the best-performing Sn–Pb PSC was 23.77% at a bias voltage of 0.76 V. It outperformed the control device, which had an SPO efficiency of 19.48%. More broadly, P-based cations have demonstrated general applicability for enhancing device performance. Moreover, BTPCI has been identified as the optimal additive compared with its analogues and Br-based salts (Supplementary Figs. 40–43). The improved performance of BTPCI-based devices can be mainly attributed to a reduced defect density, suppressed non-radiative recombination losses and enhanced interfacial carrier transport (Supplementary Figs. 44–49 and Supplementary Table 4).

To assess the operational stability of the PSCs, unencapsulated devices were subjected to MPP tracking under 1-sun irradiation in an N_2 -filled glovebox at approximately $55^\circ C$. As shown in Fig. 3f, the PCE of the control device declined to 69% of its initial value after 202 h, whereas the BTPCI device maintained 78% of its initial PCE after 572 h under the same conditions. In addition, both shelf-storage stability and thermal and humidity stability of the unencapsulated devices were substantially improved (Fig. 3g and Supplementary Figs. 50 and 51).

All-perovskite TSCs

Encouraged by the significantly improved performance of single-junction 1.25-eV-bandgap Sn–Pb PSCs, we incorporated them as the bottom subcells in all-perovskite TSCs, pairing them with 1.77 eV wide-bandgap (WBG) perovskite top subcells composed of $FA_{0.8}Cs_{0.2}Pb(I_{0.6}Br_{0.4})_3$. The thicknesses of the WBG and NBG perovskite absorption layers were optimized to 420 nm and 1,050 nm, respectively, to achieve a well-matched current density between the subcells (Fig. 4a). The photovoltaic parameters of the single-junction WBG PSC are summarized in Supplementary Table 5. It had a PCE of 20.34%, $V_{oc} = 1.35$ V, $J_{sc} = 18.39\text{ mA cm}^{-2}$ and an FF of 82.16% (Supplementary Fig. 52) when measured under a reverse voltage scan. We then fabricated all-perovskite TSCs with and without BTPCI. The statistical distribution of PCEs for both control and BTPCI-treated TSCs is shown in Fig. 4b. Notably, we achieved a V_{oc} of up to 2.224 V (Supplementary Fig. 53 and Supplementary Table 6), one of the highest values reported for two-terminal all-perovskite TSCs (Supplementary Table 7). The $J-V$ curves of the best-performing BTPCI-treated TSC exhibited a PCE of 30.17% (29.98%), $V_{oc} = 2.19$ V (2.18 V), $J_{sc} = 16.49\text{ mA cm}^{-2}$ (16.56 $mA\ cm^{-2}$) and an FF of 83.52% (82.91%) under a reverse (forward) scan (Fig. 4c and Supplementary Table 8). Independent certification further confirmed the device performance, yielding a PCE of 29.18% for a representative device (Supplementary Fig. 54). The EQE spectra of the best-performing

all-perovskite TSC (Fig. 4d) showed that the integrated J_{sc} values for the WBG and NBG subcells were 16.48 mA cm⁻² and 16.27 mA cm⁻², respectively. The SPO efficiency of the best-performing all-perovskite TSC was approximately 29.76% (Fig. 4e).

To evaluate the operational stability of all-perovskite TSCs, we conducted ageing tests by subjecting encapsulated devices to MPP tracking under 1-sun illumination in an N₂-filled glovebox at approximately 55 °C. Under these conditions, the BTPCI-treated all-perovskite TSC retained 90.02% of its initial PCE after 2,088 h (Fig. 4f). This is particularly notable, as nearly all previously reported all-perovskite tandems with PCE > 28% exhibited substantial degradation within 1,000 h (Fig. 4g and Supplementary Table 9).

Expanding the scope of 1D trimethylphosphonium perovskitoids in optoelectronic devices

Beyond 1.25-eV-bandgap Sn–Pb PSCs, the 1D trimethylphosphonium perovskitoids can significantly improve the performance of PSCs across various bandgaps and even be used in perovskite light-emitting diodes. To demonstrate this, we fabricated an inverted 1.77-eV WBG PSC with the structure ITO/NiO_x/[4-(3,6-dimethyl-9H-carbazol-9-yl)butyl]phosphonic acid (Me-4PACZ)/FA_{0.8}CS_{0.2}Pb(I_{0.6}Br_{0.4})₃/C₆₀/BCP/Cu. As shown in Supplementary Fig. 55, incorporating BTPCI led to an increase in V_{oc} from 1.27 V to 1.31 V, an improvement in J_{sc} from 18.03 mA cm⁻² to 18.34 mA cm⁻² and an increase in the FF from 83.84% to 84.04%, compared with the control group. Consequently, the SPO PCE improved from 19.05% to 20.09%.

Inspired by the exceptional properties of 1D trimethylphosphonium perovskitoid-incorporated perovskites, we further extended this strategy to fabricate inverted pure green PeLEDs with the structure ITO/polyvinylcarbazole/perovskite/2,2',2''-(1,3,5-benzinetriyl)-tris(1-phenyl-1H-benzimidazole) Li/Al. LD perovskites are typically required to optimize device performance (Fig. 4h). A statistical analysis of ten devices per condition (Fig. 4i) revealed that the highest average EQE was achieved at a BTPCI concentration of 1 mg ml⁻¹, having increased from 23.5% in the control group to 30.7% in the BTPCI-treated devices. The J – V – L characteristics of PeLEDs with and without BTPCI are shown in Supplementary Fig. 56. Notably, the brightness of BTPCI-modified PeLEDs was 1.3 times higher than that of the control devices. These results highlight the broad impact of 1D trimethylphosphonium perovskitoids in enhancing the performance of perovskite-based optoelectronic devices.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41566-026-01972-6>.

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Guang Li^{1,2,7}, Wei Ai^{1,2,7}, Weiwei Meng^{3,7}, Anamika Mishra^{4,7}, Shiqiang Fu¹, Guoyi Chen¹, Fang Yao^{1,5}, Wenbo Li¹, Shun Zhou¹, Zhiqiu Yu¹, Haibing Wang¹, Jiahao Wang^{1,6}, Shining Zhang¹, Dexin Pu¹, Senke Cheng¹, Yunfei Xiong¹, Yong Liu¹, Ying Zhang¹, Ti Wang¹, Zhenhua Yu¹ & Weijun Ke^{1,2,5}✉

¹Key Lab of Artificial Micro- and Nano-Structures of Ministry of Education, School of Physics and Technology, Wuhan University, Wuhan, People's Republic of China. ²Wuhan University Shenzhen Research Institute, Shenzhen, People's Republic of China. ³College of Engineering, Huazhong Agricultural University, Wuhan, People's Republic of China. ⁴Photovoltaic Metrology Section, Advanced Materials and Device Metrology Division, CSIR-National Physical Laboratory, New Delhi, India. ⁵State Key Laboratory of New Textile Materials and Advanced Processing, Wuhan Textile University, Wuhan, People's Republic of China. ⁶School of New Energy, Ningbo University of Technology, Ningbo, People's Republic of China. ⁷These authors contributed equally: Guang Li, Wei Ai, Weiwei Meng, Anamika Mishra. ✉e-mail: weijun.ke@whu.edu.cn

Methods

Materials

Tin iodide (SnI_2), lead thiocyanate ($\text{Pb}(\text{SCN})_2$, 99.5%), tin fluoride (SnF_2 , 99%), formamidinium iodide (FAI, 99.9%), methylammonium iodide (MAI, 99.9%) and lead bromide (PbBr_2 , 99%) were purchased from Advanced Election Technology Co., Ltd. Phenylethylammonium bromide (PEABr, 99.9%), phenylethylammonium chloride (PEACl, 99.9%), amantadine, caesium bromide (CsBr , 99.999%), poly(*n*-vinyl carbazole) (PVK), lithium fluoride (LiF, >99.99%), *N,N*-dimethylmethanamide (DMF, anhydrous), 1,4,7,10,13,16-hexaoxacyclooctadecane (18-crown-6, 99%), dimethyl sulfoxide (DMSO, anhydrous), isopropanol, ethylenediammonium diiodide (EDA_2 , 99%) and chlorobenzene (99.8%) were purchased from Sigma-Aldrich. Poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS) was purchased from Heraeus Germany Co., Ltd. BCP (99.5%) was purchased from Jilin OLED Material Tech Co., Ltd. Biuret was purchased from Meryer. Poly(*N,N'*-bis-4-butylphenyl-*N,N'*-bisphenyl)benzidine (Poly-TPD), 1,3,5-tris(1-phenyl-1*H*-benzimidazol-2-yl)benzene (TPBi), butylammonium chloride (BACl) and fullerene (C_{60} , 99.9%) were purchased from Xi'an Polymer Light Technology. BTPCl, benzyltrimethylphosphine bromide (BTPBr) and butylphosphonium chloride (BPCl) were synthesized in-house. All commercial materials were used as received without further purification.

Preparation of hole transport layer (HTL) solutions

To prepare the HTL solution, approximately 5 mg of BTPCl was added to 1 ml of PEDOT:PSS solution. The mixture was stirred for 2 h at room temperature and subsequently filtered through a 0.45- μm filter before use.

Preparation of precursor solutions

NBG $\text{FA}_{0.7}\text{MA}_{0.3}\text{Pn}_{0.5}\text{Sn}_{0.5}\text{I}_3$ perovskites. Perovskite precursor solutions with a concentration of 2.0 M were prepared by dissolving FAI (1.4 mmol), MAI (0.6 mmol), SnI_2 (1 mmol), SnF_2 (0.1 mmol), PbI_2 (1 mmol), BTPCl (0.01 mmol) and $\text{Pb}(\text{SCN})_2$ (0.02 mmol) in 750 μl of DMF and 250 μl of DMSO. The solution was stirred for 2 h and filtered through a 0.22- μm polytetrafluoroethylene (PTFE) membrane before use.

WBG $\text{FA}_{0.8}\text{Cs}_{0.2}\text{Pb}(\text{I}_{0.6}\text{Br}_{0.4})_3$ perovskites. Perovskite precursor solutions with a concentration of 1.2 M were prepared by dissolving CsI (0.24 mmol), FAI (0.96 mmol), PbBr_2 (1.44 mmol) and PbI_2 (2.16 mmol) in 800 μl of DMF and 200 μl of DMSO. The solution was stirred for 1 h at 60 °C, followed by filtration through a 0.22- μm PTFE membrane.

Light-emitting-diode device precursors. Perovskite precursor solutions were prepared by dissolving PEABr, CsBr and PbBr_2 (0.1 mmol ml^{-1}) in a mixed solvent of DMSO and DMF with a volume ratio of 1:1 using a molar ratio of 0.4:1:1. To passivate defects, 18-crown-6 was added at a concentration of 3 mg ml^{-1} . For the BTPCl-containing perovskite precursor solutions, BTPCl was dissolved into the original precursor solutions at concentrations ranging from 0 mg ml^{-1} to 2 mg ml^{-1} . All solutions were stirred at 70 °C for 6 h and filtered through a 0.22- μm PTFE filter before use.

Device fabrication

Fabrication of single-junction NBG PSCs. Etched ITO substrates were sequentially cleaned with detergent, deionized water, acetone, isopropanol and ethanol for 15 min each. Before spin-coating, the ITO substrates underwent plasma cleaning treatment (PT13M-SE, Jiangsu LEO Science Instruments Co. Ltd). PEDOT:PSS solutions, with and without BTPCl (5 mg ml^{-1}), were spin-coated (AC200-SE, Jiangsu LEO Science Instruments Co. Ltd) at 5,000 rpm for 30 s, followed by annealing in ambient air at 140 °C for 25 min. After annealing, the substrates were transferred to an N_2 -filled glovebox with controlled humidity and oxygen levels (<0.01 ppm). A total of 45 μl of precursor solution was

spin-coated onto each substrate, first at 1,000 rpm for 10 s and then at 4,000 rpm for 40 s. At the 30-s mark of the second step, 400 μl of chlorobenzene was dropped onto the rotating substrate. The resulting perovskite films were annealed at 100 °C for 10 min. Subsequently, 70 μl of EDA_2 solution (0.5 mg ml^{-1} in isopropanol) was drop-cast and spin-coated at 5,000 rpm for 30 s, followed by annealing at 100 °C for 5 min. Finally, C_{60} (20 nm), BCP (7 nm) and Cu (100 nm) were thermally evaporated in sequence onto the NBG perovskite layer to complete the NBG PSCs using a high-vacuum system ($<5 \times 10^{-4}$ Pa)⁴⁸. The devices were measured by masking the active area (0.0948 cm^2) with a metal mask (the aperture area was 0.070225 cm^2).

Fabrication of single-junction WBG PSCs. ITO substrates were cleaned as described above. NiO_x nanoparticle solutions were spin-coated (AC200-SE, Jiangsu LEO Science Instruments Co. Ltd) at 3,000 rpm for 30 s and annealed at 130 °C in ambient air for 30 min. Me-4PACz solutions were then spin-coated onto the ITO/ NiO_x substrates for 30 s and annealed at 100 °C for 10 min. After cooling, WBG perovskite precursors were spin-coated in two steps: 2,000 rpm for 10 s followed by 6,000 rpm for 40 s. At the 30-s mark of the second step, 350 μl of diethyl ether was dropped onto the centre of the film. The films were then annealed at 65 °C for 2 min and 100 °C for 10 min. A propane-1,3-diammonium diiodide (PDAI_2) solution was spin-coated at 4,000 rpm for 30 s and annealed at 100 °C for 5 min. Finally, C_{60} (20 nm), BCP (7 nm) and Cu (100 nm) were sequentially thermally evaporated under high vacuum ($<5 \times 10^{-4}$ Pa) to complete the WBG PSCs⁴⁹. The devices were measured by masking the active area (0.0948 cm^2) with a metal mask (the aperture area was 0.070225 cm^2).

Fabrication of two-terminal all-perovskite TSCs. Following the preparation of the WBG perovskite film, a 20-nm-thick C_{60} layer was deposited by thermal evaporation. Then, a 20-nm SnO_x layer was deposited using atomic layer deposition (ALD). Afterwards, a 1-nm-thick Au layer was thermally evaporated at a rate of 0.2 $\text{\AA s}^{-1}$ under a vacuum of 2×10^{-4} Pa. Film thicknesses were monitored using a quartz crystal microbalance. The bottom subcell (NBG) was fabricated using the same process as described for the single-junction NBG PSCs, thereby completing the construction of the two-terminal all-perovskite TSCs⁵⁰. The devices were measured by masking the active area (0.0948 cm^2) with a metal mask (the aperture area was 0.070225 cm^2).

Fabrication of green light-emitting diodes. Poly-TPD films were prepared by spin-coating a solution of 10 mg of Poly-TPD in 1 ml of chlorobenzene at 4,000 rpm for 30 s, followed by annealing at 100 °C for 10 min. PVK films were formed by spin-coating a 5 mg ml^{-1} solution in chlorobenzene under the same conditions and annealing at 140 °C for 15 min. Next, [2-(9*H*-carbazol-9-yl)ethyl]phosphonic acid (2PACz) films were spin-coated from a 0.4 mg ml^{-1} solution in ethanol and annealed at 100 °C for 10 min. Perovskite films were then fabricated using a blade coating method (PF200-H, Jiangsu LEO Science Instruments Co. Ltd). The substrate was affixed to the 45 °C heated stage via vacuum suction. The blade was positioned on the left edge of the substrate with a 150- μm gap, 10 μl of perovskite precursor solution was added into the gap using a pipette and the blade was moved at a speed of 10 mm s^{-1} . After coating, an N_2 knife (flow rate 10 l min^{-1} and speed 10 mm s^{-1}) was used to rapidly dry the film. The substrates were subsequently annealed at 100 °C for 8 min. Finally, the transfer layer and electrodes were deposited in a vacuum chamber via thermal evaporation. TPBi (40 nm) and LiF/Al (1.2/60 nm) were deposited using a shadow mask under a pressure of 4.0×10^{-4} Pa. The active area of the devices was 4 mm^2 (2 mm $\times$ 2 mm).

Film and device characterization. Crystal structures and the phases of the films were characterized using an X-ray diffractometer (Bruker D8 ADVANCE) with a Cu K α radiation source. Absorption spectra were measured with a UV-vis spectrophotometer (Mini280,

Shimadzu). Depth profiling of the perovskite films was conducted using a TOF-SIMS spectrometer (Tescan Amber) at 30 kV and 250 pA. Top-view and cross-sectional scanning electron microscopy images of perovskite films and devices were acquired using a field-emission scanning electron microscope (Zeiss Sigma) and a microscope (Tescan Amber). Atomic force microscopy height images and Kelvin probe force microscopy potential distribution maps were obtained in ambient conditions using an atomic force microscope (Bruker Dimension Icon XR). X-ray photoelectron spectroscopy measurements were performed with a monochromatic Al K α X-ray source (ESCALAB 250Xi, Thermo Scientific). PL and time-resolved PL spectra were recorded using a fluorescence spectrometer (Delta Flex, Horiba) equipped with a 372-nm semiconductor laser and a time-correlated single-photon counting module. J - V characteristics were measured with a source meter (Keithley 2400) under simulated AM 1.5 G sunlight (SS-X50, Enlitech). All J - V measurements were conducted in an N₂-purged environment, with a voltage sweep rate of 20 mV s⁻¹, 15-mV incremental steps and a stabilization delay of 200 ms per data point. The spectral irradiance was calibrated to 100 mW cm⁻² using a World Photovoltaic Scale (WPVS)-certified reference solar cell (SRC-2020, Enlitech), which was calibrated against National Renewable Energy Laboratory standards. EQE spectra were recorded using a system calibrated with a standard silicon reference cell (Enley Technologies). For two-terminal all-perovskite TSCs, EQE measurements of the NBG and WBG subcells were carried out under blue light bias (~550 nm) and infrared light bias (~830 nm), respectively. Electrochemical impedance spectroscopy (EIS) plots were conducted using an electrochemical workstation (CHI 770E, Shanghai Chenhua Instruments) in the frequency range 1 MHz to 1 Hz. Mott-Schottky analyses were performed at 1 kHz with bias voltages ranging from 1.5 V to 0 V in the dark. Dark J - V measurements were conducted from -0.4 V to 0.9 V under dark conditions. The PL quantum yields were measured by a system with an integrating sphere and an excitation wavelength of 365 nm. Electroluminescence spectra, J - V - L curves and EQE spectra of PeLEDs were collected by an EQE system (LQ-100, Enli Technology) in an N₂-filled glovebox at room temperature. This system had an integrating sphere, a source meter (Keysight, B2901A) and a spectrometer. When testing devices, the bias voltages started from 0 V and increased with a step of 0.1 V. Each voltage step lasted 500 ms to ensure that the measurements were stable.

Stability testing of solar cells. Operational stability measurements were carried out under 1-sun irradiation (a light-emitting diode light source, equivalent to AM 1.5 G, 100 mW cm⁻²). The current density was continuously recorded using a source meter (eTOMMENS-6010) at a fixed bias voltage. For stability measurements, the tandem devices were encapsulated using butyl rubber and sandwiched between two glass plates, with electrical contacts extracted through solder strips. In contrast, single-junction devices were tested without encapsulation. No UV filters were applied during the testing period.

Density functional theory calculations. Our density functional theory calculations were conducted using the Vienna ab initio simulation package code^{51,52} with projector augmented wave potentials. The generalized gradient approximation of the Perdew-Burke-Ernzerhof functional was used for the exchange correlation. The cutoff energy for basis functions was 520 eV. We used Γ -only k sampling with a density of $0.02 \times 2 \text{ \AA}/\pi$. The energy convergence was within 10^{-5} eV. The pK_a values were calculated using Marvin-Sketch software (<https://www.chemaxon.com>). The Hirshfeld surface analysis was performed using the CrystalExplorer software⁵³. The decomposition energy was calculated as the total energy difference between reactants and products. The ab initio molecular dynamics simulations were performed with the CP2K package⁵⁴ at the Perdew-Burke-Ernzerhof level⁵⁵. The double- ξ valence polarization basis set⁵⁶ in combination with the norm-conserving

Goedecker-Teter-Hutter pseudopotential^{57,58} was employed by considering only the gamma point, and we set an energy cutoff of 600 Ry. Grimme's D3 correction was considered to dispose of the van der Waals interactions^{59,60}. The wavefunction was optimized using an orbital transformation method⁶¹. The supercells of (PEA)₂PbI₄ (375 atoms) and (BTP)PbI₃ (755 atoms) were run for 5 ps in the constant NVT ensemble under 500 K. The temperature control reached the expected level within 1 ps as shown in Supplementary Fig. 4. Full crystal structure snapshots for (PEA)₂PbI₄ and (BTP)PbI₃ at 1.54 ps and 5.00 ps are shown in Supplementary Fig. 5. The passivation model for the FA vacancy was constructed using a 48-formula-unit FAPbI₃ supercell. One FA cation was removed to create a vacancy, which was subsequently occupied by the adsorption of a single PEA⁺ or BTP⁺ cation at the vacant site. The slab size was specified as 25.59 Å $\times$ 25.26 Å for the (100) surface. The vacuum thickness was at least 20 Å to avoid potential interaction between adjacent layers. The bottom four atomic layers were kept fixed to mimic the bulk constraint, whereas the topmost three layers were allowed to relax. The defect passivating energy E_b was calculated using the equation $E_b = E_{\text{tot}}(\text{adsorption}) - E_{\text{tot}}(\text{slab}) - E_{\text{tot}}(\text{molecule})$, where E_{tot} represents the total energy for the corresponding system.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data are available in the main text or Supplementary Information. The data that support the findings of this study are available from the corresponding author on reasonable request. Source data are provided with this paper.

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Author contributions

G.L. fabricated the NBG and tandem devices, conducted the characterizations and helped write the paper. W.M. performed the density functional theory calculations. A.M. carried out the material synthesis. W.A. and S.F. fabricated the WBG and tandem devices. F.Y. and J.W. measured the exciton binding energy, transient photocurrent (TPC) and transient photovoltage (TPV) of the samples. G.C. and Zhiqiu Yu assisted with the fabrication of the light-emitting diode devices. W.L. performed the PL-mapping measurements. H.W. and S. Zhou conducted the PL and time-resolved PL measurements. S. Zhang, D.P., S.C., Y.X., Y.L., Y.Z., T.W. and Zhenhua Yu contributed to film fabrication and materials characterization. W.K. conceived the idea, helped write

and revised the paper, and supervised the overall project. All authors discussed the results and commented on the paper.

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Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Weijun Ke.

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► Experimental design

Please check: are the following details reported in the manuscript?

1. Dimensions

- Area of the tested solar cells ☒ Yes ☐ No Method Section, Device fabrication. "The devices were measured by masking the active area (0.0948 cm²)"
- Method used to determine the device area ☒ Yes ☐ No Method Section, Device fabrication. "The devices were measured by masking the active area (0.0948 cm²) with a metal mask (the aperture area was 0.070225 cm²)."

2. Current-voltage characterization

- Current density-voltage (J-V) plots in both forward and backward direction ☒ Yes ☐ No Fig. 3c, Fig. 4c, Fig. S52-S3, Fig. S55, Table S3, Table S5, Table S6, and Table S8.
- Voltage scan conditions ☒ Yes ☐ No Method Section, Film and device characterization. "All J-V measurements were conducted in an N₂-purged environment, with a voltage sweep rate of 20 mV s⁻¹, 15 mV incremental steps and a stabilization delay of 200 ms per data point. "
- For instance: scan direction, speed, dwell times*
- Test environment ☒ Yes ☐ No at room temperature and in an N₂-filled glovebox.
- For instance: characterization temperature, in air or in glove box*
- Protocol for preconditioning of the device before its characterization ☐ Yes ☒ No No preconditioning is required for our cell characterization
- Stability of the J-V characteristic ☒ Yes ☐ No Stable power output efficiencies near the maximum power points were provided in Fig. 3e, Fig. 4e, and Fig. S55.
- Verified with time evolution of the maximum power point or with the photocurrent at maximum power point; see [ref. 7](#) for details.*

3. Hysteresis or any other unusual behaviour

- Description of the unusual behaviour observed during the characterization ☒ Yes ☐ No Negligible hysteresis was found.
- Related experimental data ☒ Yes ☐ No J-V curves obtained from reverse and forward voltage scans were provided in Fig. 3c, Fig. 4c, Fig. S52-S3 and Fig. S55.

4. Efficiency

- External quantum efficiency (EQE) or incident photons to current efficiency (IPCE) ☒ Yes ☐ No EQE curves were provided in Fig. 3d, Fig. 4d and Fig. S55.
- A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator ☒ Yes ☐ No The integrated J_{sc} values obtained from EQE were agree well with the J_{sc} determined from the J-V measurements.
- For tandem solar cells, the bias illumination and bias voltage used for each subcell ☒ Yes ☐ No Method Section, Film and device characterization. "For 2T all-perovskite TSCs, EQE measurements of the NBG and WBG subcells were carried out under blue light (~550 nm) and infrared light (~830 nm) bias, respectively. "

5. Calibration

- Light source and reference cell or sensor used for the characterization ☒ Yes ☐ No Method Section, Film and device characterization.

Confirmation that the reference cell was calibrated and certified	☒ Yes ☐ No	Method Section, Film and device characterization. "The spectral irradiance was calibrated to 100 mW cm ⁻² using a WPVS-certified reference solar cell (SRC-2020, Enlitech), which was calibrated against National Renewable Energy Laboratory standards."
Calculation of spectral mismatch between the reference cell and the devices under test	☐ Yes ☒ No	We did not incorporate the spectral mismatch factors for the devices in our lab but the certified devices involved.
6. Mask/aperture		
Size of the mask/aperture used during testing	☒ Yes ☐ No	0.070225 cm ² .
Variation of the measured short-circuit current density with the mask/aperture area	☐ Yes ☒ No	Negligible.
7. Performance certification		
Identity of the independent certification laboratory that confirmed the photovoltaic performance	☒ Yes ☐ No	Representative solar cells were certified by Ningbo New Materials Testing and Evaluation Center.
A copy of any certificate(s) <i>Provide in Supplementary Information</i>	☒ Yes ☐ No	Fig. S54.
8. Statistics		
Number of solar cells tested	☒ Yes ☐ No	Twenty-five for narrow-bandgap and fifty for tandem devices.
Statistical analysis of the device performance	☒ Yes ☐ No	Fig. 3b, Fig. 4b, and Fig. S37.
9. Long-term stability analysis		
Type of analysis, bias conditions and environmental conditions <i>For instance: illumination type, temperature, atmosphere humidity, encapsulation method, preconditioning temperature</i>	☒ Yes ☐ No	Fig. 3f (unencapsulated, simulated 1-sun illumination, and N ₂ -filled glovebox at ~55°C), Fig. 3g (unencapsulated, simulated 1-sun illumination, and N ₂ environment at room temperature), Fig. 4f (encapsulated, simulated 1-sun illumination, and N ₂ -filled glovebox at ~55°C), Fig. S18(a reverse bias of -2 V), Fig. S19 (a reverse bias of -4 V), Fig. S50 (unencapsulated, heated using a hotplate (85°C) in an N ₂ -filled glovebox), and Fig. S51 (unencapsulated, heated using a hotplate (85°C) in a 65% humidity and ambient air environment), Method Section.